

Adaptive Algorithms for Continuous Optimization

Final Project Plan

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1 Project Topic

This project studies **adaptive first-order optimization methods** and compares them theoretically and empirically with standard non-adaptive baselines. The main focus is on how adaptive step sizes affect convergence speed, robustness to hyperparameter choice, and predictive performance in classification problems.

The project is motivated by the course topic *Adaptive Methods* and is aligned with the lecture material on Polyak step size, Adagrad, and Adam, while also extending the empirical study to RMSprop and AdaGrad-Norm. In addition, the project brief explicitly requires us to survey the theory and historical development of **cubic regularization methods** (also known as Adaptive Regularization with Cubics, or ARC), which serve as a higher-order counterpart to the adaptive first-order methods.

2 Project Goal

The goal is to produce a **survey and experimental comparison** of several adaptive optimization algorithms on two datasets. We will study their mathematical motivation, update rules, assumptions, theoretical guarantees when available, and practical behavior on convex and mildly non-convex objectives.

More precisely, we will:

1. study the theory and historical background of **cubic regularization methods**, including ARC (Nesterov & Polyak, 2006) and its stochastic variants, their mathematical foundations, and their convergence properties,
2. compare different adaptive first-order algorithms (Adam, Adagrad, AdaGrad-Norm, RMSprop) and their advantages/disadvantages,
3. evaluate them against standard baselines including SGD and a quasi-Newton method (LBFGS),
4. implement the selected adaptive methods and test them on two classification datasets.

3 Methods to be Studied

3.1 Adaptive methods

We will study the following adaptive methods:

- Polyak step size,
- Adagrad,
- Adam,
- RMSprop,
- AdaGrad-Norm.

3.2 Higher-order / curvature-based methods (theory only)

As required by Objective 1 of the project brief, we will cover the following in the report’s theory section (no implementation required):

- **Cubic Regularization** (Nesterov & Polyak, 2006): minimizes a cubic upper bound on the objective at each step, achieving $\mathcal{O}(\epsilon^{-3/2})$ complexity to an ϵ -first-order stationary point,
- **ARC** (Adaptive Regularization with Cubics, Cartis et al. 2011): practical variant with adaptive cubic regularization parameter,
- brief discussion of stochastic cubic variants and their connection to adaptive first-order methods.

3.3 Baselines

To make the comparison meaningful, we will include:

- Gradient Descent / Stochastic Gradient Descent (SGD),
- one quasi-Newton baseline: **L-BFGS**.

Why this choice. SGD provides the standard first-order baseline, while L-BFGS provides a curvature-based baseline with different per-iteration cost and different practical behavior. This makes the study better aligned with the stated project objective of comparing adaptive methods with other optimization approaches.

4 Optimization Problems

We will test the algorithms on the following objectives.

4.1 Convex objective: logistic regression

Given data $(x_i, y_i)_{i=1}^n$ with $x_i \in \mathbb{R}^d$ and $y_i \in \{-1, 1\}$, we consider

$$f(w) = \frac{1}{n} \sum_{i=1}^n \log(1 + \exp(-y_i \langle w, x_i \rangle)).$$

This objective is convex and smooth.

4.2 Mildly non-convex objective

We will also consider logistic regression with a non-convex regularizer:

$$F(w) = f(w) + r(w), \quad r(w) = \lambda \sum_{j=1}^d \frac{\alpha w_j^2}{1 + \alpha w_j^2}.$$

This allows us to examine how adaptive methods behave beyond the purely convex setting.

5 Datasets

We will use two datasets:

- **A9a** (binary classification),
- **MNIST** (multi-class classification).

Rationale. A9a gives a standard sparse binary classification benchmark, while MNIST provides a larger multi-class problem. Together they allow us to compare performance across two different data regimes.

6 Theory Section of the Report

The report will include a concise but rigorous theory section covering:

- **cubic regularization:** historical development (Nesterov & Polyak 2006, ARC by Cartis et al. 2011), the cubic subproblem, convergence rates, and comparison with first-order methods,
- motivation for adaptive step sizes in first-order methods,
- formal update rules for each adaptive method,
- assumptions under which convergence guarantees hold,
- differences between convex, strongly convex, stochastic, and non-convex settings,
- caveats of Adam and why convergence guarantees are more delicate,
- conceptual comparison between coordinate-wise adaptation and scalar adaptation.

Course alignment. Special emphasis will be placed on the methods explicitly covered in the lecture notes: Polyak step size, Adagrad, and Adam. RMSprop and AdaGrad-Norm will be presented as additional methods included in the project survey. Cubic regularization is treated as a background / motivation section required by the project brief.

7 Experimental Questions

Our experiments will address the following questions:

- Q1. Do adaptive methods decrease the objective faster than SGD in practice?
- Q2. Are adaptive methods less sensitive to the initial learning rate?
- Q3. How do adaptive methods compare with L-BFGS in terms of speed, stability, and final accuracy?
- Q4. Does the ranking of methods change between the convex and non-convex objectives?
- Q5. Are some methods particularly effective on sparse features or noisy gradients?

8 Evaluation Protocol

For each method, we will report:

- training objective versus iteration,
- training objective versus wall-clock time,
- test accuracy or classification error,
- sensitivity to hyperparameters,
- robustness across the two datasets.

Whenever possible, we will keep the experimental setup consistent across methods, including stopping criteria, initialization, preprocessing, and regularization choices.

9 Implementation Plan

The project repository will contain:

- optimizer implementations,
- dataset loaders and preprocessing utilities,
- objective and gradient computations,
- one main notebook or script for reproducible experiments,
- plotting utilities for figures used in the report and presentation.

Implementation note. For L-BFGS, we may rely on a standard scientific computing implementation rather than coding the full method from scratch, since the focus of the project is the comparison rather than re-deriving every solver implementation detail.

10 Division of Work

We divide the work by coherent blocks rather than splitting optimizers arbitrarily.

10.1 Teammate 1: Theory and baselines

- write the **cubic regularization theory section**: historical development, cubic subproblem, convergence guarantees (Nesterov & Polyak 2006; ARC, Cartis et al. 2011),
- write the background sections for Polyak step size, Adagrad, and Adam from the lecture notes,
- implement or organize the SGD baseline,
- set up the L-BFGS baseline,
- help interpret results from a theoretical optimization perspective.

10.2 Teammate 2: Extended adaptive methods and experiments

- implement RMSprop and AdaGrad-Norm,
- handle dataset loading and preprocessing,
- run the experiment pipeline,
- generate plots and tables,
- write the experimental setup and results sections.

10.3 Shared responsibilities

Both teammates will jointly:

- decide final hyperparameter grids,
- validate correctness of implementations,
- discuss and interpret results,
- prepare the final slides,
- rehearse the presentation,
- revise and polish the final report.

11 Timeline

Date	Milestone
30.04	Finalize scope, collect references, download datasets, clean repository structure.
07.05	Implement all selected methods and verify gradients / updates on small tests.
14.05	Complete first round of experiments and generate preliminary plots.
19.05	Finalize slides, key tables, and presentation storyline; rehearse jointly.
20.05	In-class presentation.
21.07	Submit final report and recorded video.

12 Expected Structure of the Final Report

The final report will have the following structure:

1. Introduction and motivation,
2. Background: cubic regularization methods (theory, historical development),
3. Background: adaptive first-order methods,
4. Problem setup and objectives,
5. Algorithms and theoretical comparison,
6. Experimental setup,
7. Results,
8. Discussion,
9. Conclusion.

13 Expected Contributions

By the end of the project, we expect to provide:

- a clear comparison of several adaptive methods,
- a course-aligned explanation of their theoretical foundations,
- an empirical study on two datasets,
- practical insight into when adaptive methods help and when they do not.

14 References

The following references will form the starting point of the project bibliography:

- Diederik P. Kingma and Jimmy Ba. *Adam: A Method for Stochastic Optimization*. 2014.
- Rachel Ward, Xiaoxia Wu, and Leon Bottou. *Adagrad Stepsizes: Sharp Convergence over Nonconvex Landscapes*. 2020.
- Sebastian Ruder. *An Overview of Gradient Descent Optimization Algorithms*. 2016.
- Yurii Nesterov and Boris T. Polyak. *Cubic regularization of Newton method and its global performance*. Mathematical Programming, 108(1):177–205, 2006.
- Coralia Cartis, Nicholas I. M. Gould, and Philippe L. Toint. *Adaptive cubic regularisation methods for unconstrained optimization. Part I: motivation, convergence and numerical results*. Mathematical Programming, 127(2):245–295, 2011.

Additional references may be added for RMSprop, AdaGrad-Norm, L-BFGS, and stochastic cubic regularization.

Final Note

This plan is designed to cover the full project brief: theory, comparison of adaptive methods, comparison against non-adaptive and quasi-Newton baselines, and implementation on two datasets.